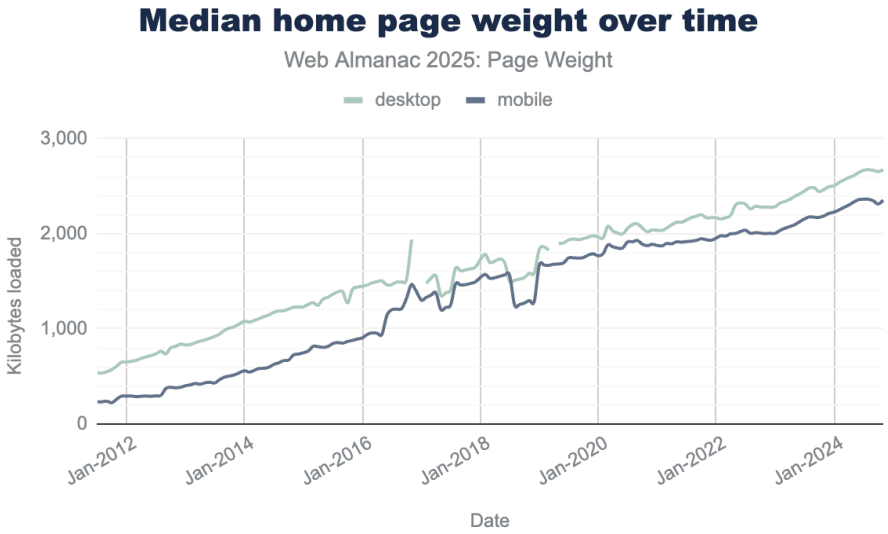
Teaching the Next Web

From hybrid rendering to agentic hypermedia

Juho Vepsäläinen

6.5.2026

Why this matters now



Source: HTTP Archive Web Almanac 2025, Page Weight, Fig. 14.1

Why this matters now

The web is increasingly shaped by software agents.

Search and click

Users searched, opened links, and did the synthesis themselves. [6]

Agentic search

Systems now browse and compare sources on the user's behalf. [7], [8]

Browser use

Websites now increasingly serve agents as well as people. [9], [10]

Outline

Part 1	Research highlights
Part 2	From research to teaching
Part 3	Future vision and goals
Part 4	Conclusion

Part 1

Research highlights

Hybrid rendering models

Dissertation: Vepsäläinen, J. (2025). Emergence of hybrid rendering models in web application development.

Dissertation background

From static content to web applications.

The platform was designed for static content, then evolved into an application platform.

Choose rendering strategies case by case.

Application size tripled in a decade while mobile constraints remained.

Main claims

Rendering should follow use cases.

The mainstream SPA model should be used more sparingly.

Hybrid rendering narrows dynamic behavior to where it is needed.

Islands constrain dynamic scope. Resumability avoids hydration and enables code-splitting.

Simplification agenda

How do we reduce unnecessary complexity without giving up capability?

Revisiting Hypermedia, The Forgotten Web Application Development Paradigm

Vepsäläinen • TechRxiv • 2026

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The Rise of Disappearing Frameworks in Web Development

Juho Vepsäläinen, Arto Hellas, Petri Vuorimaa • ICWE 2023

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The Case for HTML First Web Development

Vepsäläinen • JWE (accepted) • 2026

Part 2

From research to teaching

AI and computing education

Educational implications

- **Curriculum and pedagogy need rethinking.**

What changes in practice

Assessment and classroom design

Flipped learning, oral exams, dialogical assessment.

[1] Vepsäläinen, J., & Juntunen, P. (2026). Artificial intelligence for computer science education - hype or help?

AI and computing education

Educational implications

- **Curriculum and pedagogy need rethinking.**
- **Integrity is not the only issue.**

What changes in practice

Assessment and classroom design

Flipped learning, oral exams, dialogical assessment.

Beyond integrity

Also trust, over-reliance, bias, and unequal access.

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AI and computing education

Educational implications

- **Curriculum and pedagogy need rethinking.**
- **Integrity is not the only issue.**
- **The goal is responsible technical judgment.**

What changes in practice

Assessment and classroom design

Flipped learning, oral exams, dialogical assessment.

Beyond integrity

Also trust, over-reliance, bias, and unequal access.

Durable skills

Critical reading, writing, debugging, reflection.

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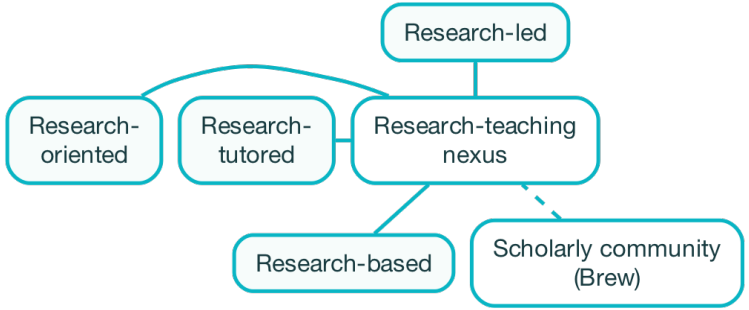
How research informs my teaching

- **Build working models students can adapt.**
  Research helps me teach principles, not only tools.

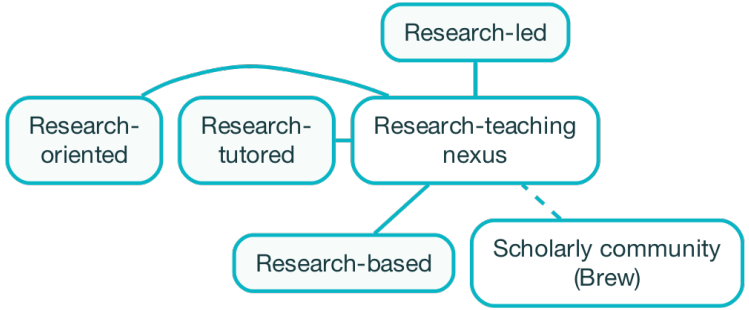


Research-teaching nexus [3,4]

How research informs my teaching

- **Build working models students can adapt.**
  Research helps me teach principles, not only tools.

- **Teaching and supervision shape my questions.**
  Student work reveals topics that deserve more attention.



Research-teaching nexus [3,4]

Research with students

20+
BSc theses since 2022

6
MSc theses advised

Current advising work covers agents and MCP, hypermedia, microfrontends, cloud migration, and WebAssembly.

- **Pyry Pohjalainen (2025): AI for web development; AAAI Magazine.**
- **Anoop Bidikar (2024): Stateful serverless architectures at the edge.**
- **Antti Nousiainen (2024): WebAssembly in edge computing.**

Writing support

A writing process students can use

- **Shared writing process improves supervision.**
  "The Process of Scientific Writing" makes publishing expectations concrete.

- **Students can use it on their own.**
  It supports supervision through a shared writing process.

SURVIVEJS

The
Process of
Scientific Writing

Your guide to getting started with scientific publishing

Juho Vepsäläinen

Public learning communities

Teaching also happens outside courses

- **I organize Future Frontend.**

  Meetups and the conference keep me grounded in current practice.

- **My developer background supports my teaching.**

  webpack-merge is one example; earlier work includes webpack and Blender.

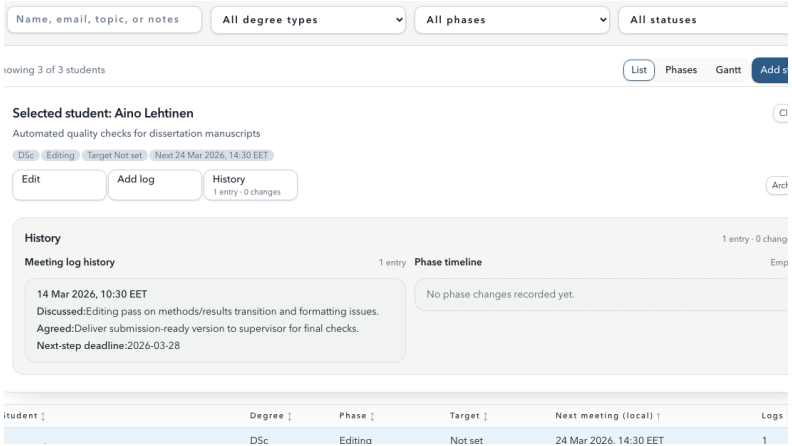
Teaching outside the classroom

Future Frontend is one way I do it.

Tools that support my work with students

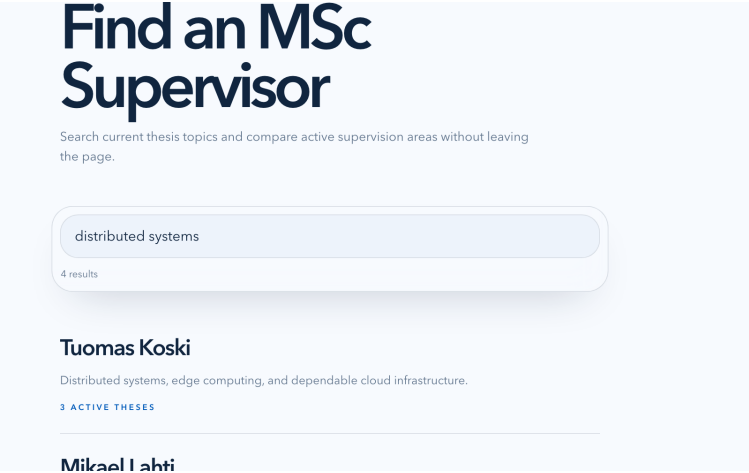
Thesis Journey Tracker

Phases, meetings, and notes in one place.



Supervisor Search

Find suitable MSc supervisors without leaving the page.



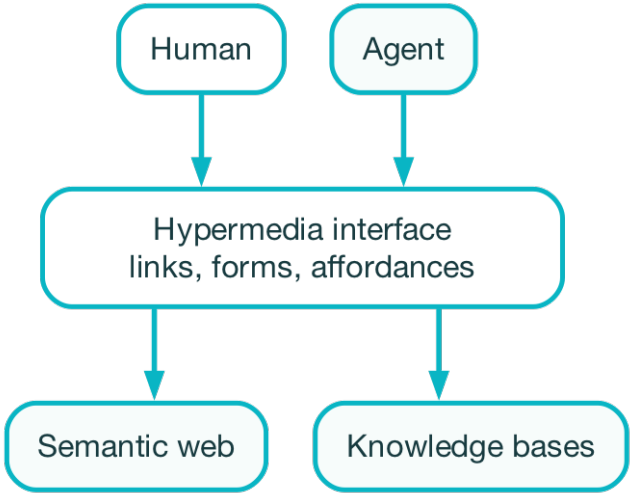
Part 3

Future vision and goals

Future vision: agentic hypermedia

Research questions

- How do we keep the web usable by agents?
- How do semantic web ideas and knowledge bases support agent-facing systems?
- What should remain stable in web architecture as agents become users?



[2], [5] Fielding; Berners-Lee et al.

My contribution to Aalto

- I would strengthen Aalto's profile in web architecture, agent-facing systems, and modern software practice.

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My contribution to Aalto

- I would strengthen Aalto's profile in web architecture, agent-facing systems, and modern software practice.
- I already build research through student work in my topic areas.
- I build infrastructure and communities that make this work visible, usable, and connected to practice.

Part 4

Conclusion

Conclusion

I bring a distinctive specialization in web architecture, agent-facing systems, and modern software practice.

Q&A

References

[1] Vepsäläinen, J., & Juntunen, P. (2026). Artificial intelligence for computer science education - hype or help? Preprint.
www.researchgate.net/publication/400580916_Artificial_intelligence_for_web_development_Perspectives_from_the_industry

[2] Fielding, R. T. (2000). Architectural Styles and the Design of Network-based Software Architectures. PhD dissertation, UC Irvine.
ics.uci.edu/~fielding/pubs/dissertation/

[3] Healey, M. (2005). Linking research and teaching to benefit student learning. Journal of Geography in Higher Education, 29(2), 183-201.
doi.org/10.1080/03098260500130387

[4] Brew, A. (2012). Teaching and research: new relationships and inquiry-based higher education. Higher Education Research & Development, 31(1), 101-114.
doi.org/10.1080/07294360.2012.642844

[5] Berners-Lee, T., Hendler, J., & Lassila, O. (2001). The Semantic Web. Scientific American, 284(5), 34-43.
lassila.org/publications/2001/SciAm.html

[6] Gou, B., Huang, Z., Ning, Y., et al. (2025). Mind2Web 2: Evaluating Agentic Search with Agent-as-a-Judge. NeurIPS Datasets & Benchmarks.
osu-nlp-group.github.io/Mind2Web-2/

[7] OpenAI. (2025). Introducing deep research.
openai.com/index/introducing-deep-research/

[8] OpenAI. (2025). Introducing Operator.
openai.com/index/introducing-operator/

[9] Anthropic. (2024). Introducing computer use, a new Claude 3.5 Sonnet, and Claude 3.5 Haiku.
www.anthropic.com/news/3-5-models-and-computer-use

[10] Google. (2024). Google introduces Gemini 2.0: A new AI model for the agentic era.
blog.google/innovation-and-ai/models-and-research/google-deepmind/google-gemini-ai-update-december-2024/